

The population generative model

1 What the model does

The model gives a distribution of parameters for a population of PDAC patients. One draw from this distribution is one patient. The model then predicts what a published study of these patients shows. The fit compares this prediction with the numbers in the papers.

We never observe one patient. We observe a median and an interquartile range from a cohort. The model must therefore run all the way from a population of parameters to a cohort summary before it can compare anything.

2 The terms

Throughout, $\odot$ is an elementwise product and $\text{diag}(v)$ is the diagonal matrix with v on its diagonal. A logarithm of a vector acts elementwise.

Each table gives the status of the term. **inferred** means that the fit estimates it. **stage 1** means that we compute it from the stage-1 output before the fit. **input** means that we set or build it before the fit by some other route. **derived** means that it is a function of the other terms. **data** means that we read it from a paper.

Sizes

All sizes are constants.

Term	Value	Meaning
P	271	The number of model parameters.
M	65	The number of observables, over all scenarios.
M_w	49	The number of observables whose source reports a usable scale.
N	2×10^4	The number of patients that we simulate. Section 6 sets it.
C	29	The number of study cohorts. The pooled CD8 target splits into three; see section 8.5.
A	118	The number of anchor rows. It is 67 centre rows plus 51 width rows, because one observable now has three cohorts.
n_c	6–702	The number of real patients in cohort c .
B	2×10^3	Bootstrap replicates for the sampling covariance.
B'	200	Independent clouds for the Monte Carlo covariance.

Earlier versions of this note wrote R for both the correlation matrix and the number of anchor rows. The anchor count is now A . R is only the matrix.

Quantities for one patient

Term	Size	Status	Meaning
θ_i	P	derived	The parameter vector of patient i . All values are positive.
$\vartheta_i = \log \theta_i$	P	derived	The same vector on the log scale.
z_i	P	input	The latent coordinates of patient i .
x_i	M	derived	The observables of patient i , on the log scale.

Quantities for the population

Term	Size	Status	Meaning
μ	P	inferred	The centre of the population, on the log scale.
ω	P	derived	The spread of each parameter between patients, on the log scale.
s	1	inferred	The global width multiplier, on the log scale. It scales every assumed width at once.
u	P	inferred	The departure of each assumed width from the global level. It sums to zero.
$\mathcal{M}$	9	input	The set of parameters whose spread a source measured. These take no s and no u .
ω_0	P	input	The prior centre for ω . Section 9 builds it in four layers.
Ω	$P \times P$	derived	The covariance of the population, on the log scale.
R	$P \times P$	stage 1	The correlation matrix of the population. Section 10.
L_R	$P \times P$	input	The Cholesky factor of R , so that $R = L_R L_R^\top$.
D_ω	$P \times P$	derived	$\text{diag}(\omega)$, the matrix of marginal widths.

Quantities for the data

Term	Size	Status	Meaning
g	map	input	The model. It changes log parameters into log observables.
Q^{ctr}	M	derived	The median that the model predicts, per observable.
Q^{wid}	M_w	derived	The log interquartile range that the model predicts.
y	A	data	The same two quantities, read from the papers.
η	C	inferred	An offset for each study.
$\mathcal{E}_c$	rule	input	The eligibility rule of cohort c , where the paper reports one.
δ	M	inferred	A structural error in the centre, for each observable.
κ	M_w	inferred	A structural error in the width, for each observable with a width row.
κ_0	1	inferred	The width error shared by every observable.
ε	M	input	The held-out residual of the emulator, for each observable.
Σ	$A \times A$	input	The covariance of the observed rows. The fit does not change it.
Σ_{MC}	$A \times A$	input	The part of Σ that comes from the finite cloud. It is common to every cohort.

Quantities that control the priors

Term	Value	Status	Meaning
μ_0	P	stage 1	The centre of the prior on μ . Section 8.1.
Σ_1	$P \times P$	stage 1	The covariance of the prior on μ . The stage-1 covariance on the log scale.
τ_s	0.5	input	The width of the prior on s . It trades directly against κ_0 .
τ_ω	0.5	input	The width of the prior on u , on the log scale.
$\tau_{\omega,j}$	—	input	The width of the prior on $\log \omega_j$ for $j \in \mathcal{M}$. It comes from the sample size n_j .
τ_η	—	inferred	The scale of the study offsets.
τ_δ	—	inferred	The scale of the centre errors.
τ_κ	—	inferred	The scale of the width errors.
ϕ	—	inferred	All the quantities that the fit estimates.

The full parameter vector is

$$\phi = (\mu, u_{\text{raw}}, s, \log \omega_{\mathcal{M}}, \eta_{\text{raw}}, \delta_{\text{raw}}, \kappa_{\text{raw}}, \log \kappa_0, \log \tau_\eta, \log \tau_\delta, \log \tau_\kappa).$$

3 How to make one patient

1. Draw z_i from a standard normal distribution. Do this one time only, before the fit.
2. Multiply by L_R . This step installs the correlations.
3. Multiply element j of the result by ω_j . This step sets the width of each parameter.
4. Add μ .
5. The result is ϑ_i .
6. Take the exponential of each element. The result is θ_i .

In one equation:

$$\vartheta_i = \mu + D_\omega L_R z_i, \quad z_i \sim \mathcal{N}(0, I_P), \quad \theta_i = \exp(\vartheta_i). \quad (1)$$

One cloud serves every cohort. Cohorts differ only by the eligibility rule $\mathcal{E}_c$ of section 8.4, which selects a subset of that one cloud. This is deliberate twice over. It states that every study samples the same population, which is the claim the model is making. It also makes the Monte Carlo error common to every cohort, so section 6 can model that error as one object rather than as C independent ones.

The covariance of the population follows directly:

$$\vartheta_i \sim \mathcal{N}(\mu, \Omega), \quad \Omega = D_\omega L_R L_R^\top D_\omega = D_\omega R D_\omega. \quad (2)$$

This form separates the two halves of a covariance matrix. R holds the correlations and the fit does not change it. ω holds the marginal widths and the fit estimates all P of them. The construction is known as the separation strategy.

Two properties follow, and both matter for interpretation.

$$\sqrt{\Omega_{jj}} = \omega_j \quad \omega_j \text{ is exactly the width of parameter } j. \quad (3)$$

$$\text{Corr}(\Omega) = R \quad \text{The correlations do not move when the fit moves } \omega. \quad (4)$$

A diagonal rescaling cannot change a correlation matrix, so the second property holds at every value of ω , not only at one point.

Why step 1 happens only one time. We draw z once and we keep it. The map from ϕ to $Q(\phi)$ is then continuous and has no random noise. The sampler needs this property to calculate a gradient. The same N patients move continuously as ϕ moves.

The map is continuous but it is not smooth. A quantile selects an order statistic, and the selected patient changes when two patients swap rank. An eligibility rule has the same defect at its threshold. Both give a piecewise gradient, and only the few patients near each level carry it. We therefore use a smooth quantile: weight patient i by a kernel in its own rank, and take a weighted quantile. `inference/importance.py` already supplies `weighted_quantile`. The bandwidth is a numerical choice and it must be small against the width of the cloud.

The remaining cost is a fixed approximation error of order $N^{-1/2}$ in Q . Section 6 measures that error and adds it to Σ .

4 How to predict the data

1. Make N patients with the steps in the previous section, one cloud per cohort.
2. Put each patient through the model g . This gives N observable vectors $x_i = g(\vartheta_i)$.
3. Where the paper reports an eligibility rule, keep only the patients that satisfy it. Use a smooth weight at the threshold, not a hard indicator.
4. For each observable, take the smooth weighted quantiles at 0.25, 0.50 and 0.75.
5. Reduce them to two rows.

The two rows are the centre and the log width:

$$Q_m^{\text{ctr}}(\phi) = \hat{F}_{m,\phi}^{-1}(0.5), \quad Q_m^{\text{wid}}(\phi) = \log \{ \hat{F}_{m,\phi}^{-1}(0.75) - \hat{F}_{m,\phi}^{-1}(0.25) \}. \quad (5)$$

An observable whose source reports a centre only has a centre row and no width row. There are $M = 65$ observables with a centre and $M_w = 49$ with a width. One observable carries three cohorts, so the rows number 67 and 51, and $A = 118$. Section 8.5 explains the difference between an observable and a row.

Why the rows are the centre and the log width. An earlier version used the three raw quantiles as three rows. Three changes follow from this one.

The sampling distribution of an interquartile range is skewed and is bounded below by zero. At $n_c = 6$ a normal error term on the raw quantiles is wrong. On the log scale the same quantity is close to normal at every cohort size that we have.

The width correction becomes additive. Section 6 shows this. An additive correction on one row makes the aliasing of section 8.6 visible instead of implicit.

The count of width parameters falls from 65 to 49. Sixteen observables have no width row, so their κ_m was never identified and always returned its prior. Those entries are now absent rather than reported.

There is no closed form for $F_{m,\phi}$. The model g is a set of differential equations and it is not linear, so the pushforward of a log-normal through it belongs to no standard family. Step 4 is therefore a numerical calculation, not a shortcut. In practice g is a neural-network surrogate for the simulator, which is what makes $C \times N$ evaluations per gradient step affordable.

5 What the emulator must cover

The emulator sets a hard limit on what the fit can say, and the limit is a property of the training pool and not of the sampler.

Every parameter varies in the pool. If the pool held a parameter fixed, the emulator never saw that parameter move. The likelihood along that direction is then not flat. It does not exist, and a θ that varies the parameter is an extrapolation. The pool must therefore vary all P widths that the fit estimates.

The pool must cover the flanks. The rows in (5) read the cloud at 0.25 and 0.75, not at the centre. A design that covers the centre of the prior well and the tails poorly gives its worst predictions exactly where the data enter.

Measure ε where the fit reads the emulator. The held-out residual of the training design is the wrong number to use. After the fit, draw from $F(\theta \mid \hat{\phi})$, run those draws through the full simulator, and compare. If the error there exceeds the design error, the fit read the emulator outside its region and the run is not valid.

Report a resolution number per parameter. With P active inputs, a direction can look flat because the emulator did not resolve it. That reads the same as a flat likelihood and it is a different thing. A per-parameter sensitivity of the emulator, measured on the pool, separates the two.

6 The observation equation

The model corrects its prediction before the comparison. One correction moves the centre. The other changes the width.

$$\tilde{Q}_m^{\text{ctr}} = Q_m^{\text{ctr}}(\phi) + \eta_{c(m)} + \delta_m, \quad (6)$$

$$\tilde{Q}_m^{\text{wid}} = Q_m^{\text{wid}}(\phi) + \log \kappa_m. \quad (7)$$

Then

$$y = \tilde{Q}(\phi) + e, \quad e \sim \mathcal{N}(0, \Sigma). \quad (8)$$

Each term answers one question about how the model can be wrong.

η_c absorbs a measurement offset that applies to a whole study. δ_m absorbs a systematic error of the model in the centre of observable m . κ_m absorbs a systematic error of the model in the *width* of observable m , so $\kappa_m = 1$ means no error and $\kappa_m = 2$ means the model produces half the spread that the study reports.

Without κ the model has no way to be wrong about a width. Only ω could then close a gap in spread, and the fit would report a fault of the simulator as diversity between patients. Section 8.6 explains what separates the two.

6.1 How we build Σ

Σ has three parts. Each part comes from a different cause.

$$\Sigma = \Sigma_{\text{samp}} + \Sigma_{\text{MC}} + \text{diag}(\varepsilon^2). \quad (9)$$

Σ_{samp} : the cohort has a finite number of patients. We do not use an asymptotic formula. We resample the model's own cloud:

1. Take the cloud at the plug-in value ϕ_0 .
2. Draw n_c patients from it with replacement.
3. Compute the rows of cohort c from those n_c patients, with the same quantile convention that the paper used.
4. Repeat B times. Take the sample covariance across the rows of that cohort.

This is exact at every n_c . It needs no density estimate and no bandwidth. It carries the dependence between observables of the same cohort exactly, because the cloud carries it. Rows of different cohorts get zero entries here, because the real patients differ. Σ_{samp} is therefore block diagonal by cohort.

The formula, if you want one. For a sample of size n from any continuous distribution F , the order statistic $X_{(k)}$ satisfies $F(X_{(k)}) \sim \text{Beta}(k, n+1-k)$ exactly, at every n . Two levels are jointly Dirichlet. So you can draw $u \sim \text{Beta}(k, n+1-k)$, map through the cloud's own quantile function $\hat{F}_{m, \phi_0}^{-1}$, and read the moments off the result. This is distribution free and it needs no asymptotics.

The bootstrap above is the same calculation, and it also reproduces whatever interpolation rule the paper applied. We prefer it for that reason.

The earlier version of this note used the Bahadur covariance, which needs a density estimate f_m at each quantile and a bandwidth $h = 0.035$. That formula is an asymptotic approximation to the quantity above. It is poor at $n_c = 6$, and some cohorts are that small.

Σ_{MC} : the cloud is finite. One cloud of N patients makes the prediction for all C cohorts. Its error is therefore the same error in every row, and not an independent error per cohort. We measure it directly. Draw B' independent clouds of size N at ϕ_0 . Compute the full A -vector $\tilde{Q}$ for each cloud. Take the sample covariance across the B' clouds. The result is dense across cohorts. Keep it as a low-rank matrix plus a diagonal when $B' < A$.

Report $\text{tr } \Sigma_{\text{MC}} / \text{tr } \Sigma_{\text{samp}}$. A reader must be able to see whether the cloud mattered.

Reduce this term before you model it. Two actions do most of the work. Raise N : the term falls as N^{-1} and the emulator makes N cheap. At $N = 2 \times 10^4$ against the largest cohort at $n_c = 702$, it adds about 3.5%. Stratify z instead of drawing it independently: a scrambled Sobol sequence or a stratified normal sample cuts the quantile variance at the same N .

$\text{diag}(\varepsilon^2)$: the emulator is not the simulator. Section 5 says how to measure ε and where.

Σ does not change during the fit. The cloud that we resample depends on ϕ . We evaluate Σ once, at a plug-in value ϕ_0 , and then hold it fixed. The method is therefore an estimating equation, not a full likelihood.

This is deliberate. If Σ changes with ϕ , then Σ grows with the population spread, and the term $-\frac{1}{2} \log \det \Sigma$ rewards a narrow population whatever the fit quality. A test showed the size of the error. With cohorts of 20, 40 and 120 patients, a live Σ recovered 0.878, 0.941 and 0.963 of the true spread. A fixed Σ recovered 1.002, 0.999 and 1.001.

The plug-in value is not critical. A change of ϕ_0 by a factor of two moved the estimate by 0.14 to 0.32 sampling standard deviations.

Take ϕ_0 from the flat fit of section 7: the pair $(\hat{\mu}_{\text{flat}}, \omega_0)$. That fit costs about $1/N$ of this one, and it is a better anchor than the prior mean.

The frozen cloud gives a bias, not a noise. For one fixed $z_{1:N}$ the Monte Carlo error is a fixed offset in $\tilde{Q}$. It does not average away over the A rows. Σ_{MC} is correct on average over clouds, which is a hedge and not a cure.

Measure the residual bias directly. Run the whole fit three times, with three different frozen clouds, and compare the posteriors. Section 14 lists this as a required check.

7 The flat special case

The flat problem asks for one parameter vector, not a population. It fits the reported medians and it ignores the reported spreads. This model contains it, and the reduction is worth stating, because we run the flat fit first.

7.1 The reduction

Set $\omega = 0$. Then $D_\omega L_R z_i = 0$ for every i , all N patients collapse onto one point, and

$$Q_m^{\text{ctr}}(\phi) = g_m(\mu). \quad (10)$$

That is the flat forward map exactly.

The width rows do not survive the same limit. $Q_m^{\text{wid}} = \log 0$ is not defined. We remove them. The reduction is therefore a mask on the rows together with $\omega = 0$, and not a value of a parameter alone.

What remains is a small model:

$$y_{m,c}^{\text{ctr}} = g_m(\mu) + \eta_c + \delta_m + e, \quad e \sim \mathcal{N}(0, \Sigma), \quad \mu \sim \mathcal{N}(\mu_0, \Sigma_1). \quad (11)$$

	Terms
Unchanged	$\mu, \mu_0, \Sigma_1, \eta, \delta, \tau_\eta, \tau_\delta, \varepsilon$
Removed	$\omega, s, u, \omega_0, \mathcal{M}, R, L_R, \kappa, \kappa_0, \tau_\kappa, z_{1:N}, \Sigma_{MC}$
Sections that go	8.2, 8.6, 9, 10, 13
Rows	67 centre rows, against 118
Forward passes per gradient step	1, against $C \times N$

Two parts of this document carry over word for word. Section 12 on the aliasing of η and δ describes the map from observables to cohorts, and that map does not change. Section 8.5 on the CD8 split is the same, and it buys the same thing.

7.2 Three things that the mask does not do

The estimand is different, and one symbol hides it. Flat μ is the parameter vector that reproduces the reported medians. Population μ is the centre of the population. For a nonlinear g these differ, because $g(\mu) \neq Q^{\text{ctr}}(\mu, \omega)$ when $\omega \neq 0$. The difference is exactly the Jensen gap of section 8.1.

So do not carry $\hat{\mu}_{\text{flat}}$ into the population model as a centre, and do not read the population $\hat{\mu}$ as a flat calibration. Measure the gap first, as section 8.1 requires, and then translate.

Σ must come from another source. Section 6.1 resamples the model's cloud. The flat model has no cloud to resample. It has a better source: the spread that the paper itself reports. The

standard error of a reported median follows from that spread and from n_c , with no plug-in and no simulation.

Note the reversal. In the population model the interquartile range is signal. In the flat model the same number is only noise calibration. The two fits therefore use the same published table in two different ways, and a target that reports no spread is weaker in both.

The eligibility rule stops working. A filter on one point does nothing. The flat model therefore has no cohort device at all, and η_c carries every difference between studies on its own. The assumption box of section 8.4 applies to the flat model without any qualification: the assay story is asserted, and nothing in the flat fit can test it.

7.3 Why we run it first

It is a test of everything except the population layer. The prior on μ , the forward map, the corrections, the alias structure and the sampler all run in (11), at about $1/N$ of the cost. A fault in any of them is a fault that the population fit would also have. Find it in the cheap model.

It supplies the plug-in. Section 6.1 holds Σ at a value ϕ_0 . Take $\phi_0 = (\hat{\mu}_{\text{flat}}, \omega_0)$.

It measures the Jensen gap for free. Compare $g(\hat{\mu}_{\text{flat}})$ against Q^{ctr} evaluated at the same centre with ω_0 . That is the number that section 8.1 asks for, and the flat fit is where to get it.

(11) is the population code with a mask, and it is not the neural posterior estimator that answers the flat question elsewhere in this package. The two routes take the same data to the same estimand. They do not share a likelihood, a prior on the corrections, or a sampler. A disagreement between them is informative, and it is not a defect by itself.

8 The priors

8.1 The centre

$$\mu \sim \mathcal{N}(\mu_0, \Sigma_1). \quad (12)$$

μ_0 and Σ_1 are the mean and the covariance of the stage-1 composite prior, on the log scale. Where the stage-1 object is a copula, use its own `log_prob` as the prior term and skip the Gaussian approximation.

Earlier versions used $\Sigma_1 = \tau_\mu^2 I_p$ with $\tau_\mu = 0.5$. That gave a parameter constrained to within 0.05 and a parameter constrained to within 1.5 the same width, and it discarded the stage-1 correlations. Both are available at no cost, so both are now used.

The width of (12) measures how well we know the centre of the population. It shrinks as data accumulate. It is a different quantity from ω , which is the variability between patients and does not shrink. To confuse the two is the standard-error-versus-standard-deviation error, one level up in the hierarchy.

Two conditions on μ_0 , and both are checkable.

The Jensen gap. Stage 1 estimated a parameter that reproduces a reported summary. Under a nonlinear multivariate g , that parameter is not the parameter of the median patient. So $g(\mu_0)$ and $Q^{\text{ctr}}(\mu_0, \omega_0)$ differ. Measure the difference per observable and compare it against τ_δ . If it is small, record the number and continue. If it is not small, correct μ_0 by one offline solve: find the shift c with $\text{median}_i\{g(\mu_0 + c + D_{\omega_0} L_R z_i)\} = g(\mu_0)$. That costs one forward pass on the cloud.

Double use of the data. (12) is a prior on μ only if the stage-1 sources are disjoint from the C cohorts. If any publication feeds both, μ_0 counts that data twice and the posterior on μ comes out too narrow. The source lists make this checkable, and it has not yet been checked.

8.2 The spread

The widths split into two sets. A parameter is in $\mathcal{M}$ when a source measured its spread across biological units, in the sense of section 9. At present $|\mathcal{M}| = 9$.

For a measured parameter, $j \in \mathcal{M}$:

$$\log \omega_j \sim \mathcal{N}(\log \omega_{0j}, \tau_{\omega,j}^2), \quad \tau_{\omega,j}^2 = \frac{1}{2(n_j - 1) + \tau^{-2}}. \quad (13)$$

The width is the posterior width that the shrinkage (21) already implies. A measured parameter takes no global multiplier.

For an assumed parameter, $j \notin \mathcal{M}$:

$$\log \omega_j = \log \omega_{0j} + s + u_j, \quad s \sim \mathcal{N}(0, \tau_s^2), \quad u_j \sim \mathcal{N}(0, \tau_\omega^2), \quad \sum_{j \notin \mathcal{M}} u_j = 0. \quad (14)$$

We impose the constraint by centring: $u = u_{\text{raw}} - \bar{u}_{\text{raw}}$.

The split is the point. s is one number that moves every assumed width together. u holds the pattern and has $P - |\mathcal{M}| - 1$ free values. Section 8.6 needs the two to be separate quantities, because s is the one that competes with κ_0 and u is not.

The value τ_ω controls how far the data can move one width away from its anchor. On the log scale, $\tau_\omega = 0.5$ gives a central 95% interval of about $[0.37, 2.72]$ for each parameter.

This choice is not harmless. The data give at most $M_w = 49$ constraints on Ω (section 11), against $P = 271$ widths. Most parameters therefore return their prior. A large τ_ω reports a wide population that no study measured.

8.3 The corrections

$$\eta = \tau_\eta \eta_{\text{raw}}, \quad \eta_{\text{raw}} \sim \mathcal{N}(0, I_C), \quad \log \tau_\eta \sim \mathcal{N}(\log 0.15, 1), \quad (15)$$

$$\delta = \tau_\delta \delta_{\text{raw}}, \quad \delta_{\text{raw}} \sim \mathcal{N}(0, I_M), \quad \log \tau_\delta \sim \mathcal{N}(\log 0.15, 1), \quad (16)$$

$$\log \kappa = \log \kappa_0 + \tau_\kappa \kappa_{\text{raw}}, \quad \kappa_{\text{raw}} \sim \mathcal{N}(0, I_{M_w}), \quad \log \tau_\kappa \sim \mathcal{N}(\log 0.15, 1), \quad (17)$$

with $\log \kappa_0 \sim \mathcal{N}(0, \tau_s^2)$. These are non-centred forms. The centre 0.15 matches the floor of the translation-relevance rubric used at stage 1.

The prior on $\log \kappa_0$ takes the same width as the prior on s . The two compete for one direction, and giving them the same width states that we have no preference between them before the data arrive.

8.4 The cohort terms

Studies do not recruit the same patients. A cohort of resectable patients and a cohort of metastatic patients are not two samples from one distribution, so the claim that one F generates

all C cohorts is a real assumption and not a formality. This subsection says what the model does about it, and what it deliberately does not do.

All but one observable belong to exactly one cohort. Section 12 shows that a *free* per-cohort term on the observation is therefore aliased with δ , exactly, for those observables. Only two things escape that: a term that costs no parameter, or an observable that appears in more than one cohort. Section 8.5 is the second case.

The eligibility rule, where the paper reports one. Cohort c recruited patients that satisfy some criterion. Apply the criterion to the cloud and take the quantiles of the survivors. This is the model's only cohort device, and it is the right one. It adds no parameter. It is aliased with nothing. It says that the cohort is the same population under a condition, which is the literal description of what the study did. Use it wherever the criterion is quantitative and reported.

Meta-regression, once the attribute table exists. Replace the free η_c by $\eta_c = Z_c\beta$, where Z_c holds cohort attributes: stage, line of therapy, sample type, assay platform, region. The free count then falls from C to $\dim \beta$, and cohorts that share attributes share the effect. The coefficients pool across all C cohorts, which is what makes them estimable. We do not have the table yet, so η_c stays free for now.

The same device can be moved one level down, onto the parameters: $\mu_c = \mu + \Gamma z_c$, with Γ pooled across cohorts and each column tied to a declared mechanism, such as stage moving a growth rate. That is the ordinary covariate model of an NLME analysis. It is the correct upgrade path, and it is also gated on the attribute table. Declare the links one at a time and from biology. Do not fit a basis.

What η_c asserts, and what we cannot test. An additive offset on every log readout of one study is an *assay calibration* story: this platform reads high by a constant factor. A cohort-level shift in the parameters would be a *biology* story: this study recruited different patients. The two predict the same thing for a cohort with one observable, and most cohorts have about two. These data cannot separate them.

The model asserts the assay story by construction. That is a choice, it is not a finding, and any claim about μ inherits it. Section 17 lists the data that would let the two be told apart.

Why there is no cohort random effect on the parameters. An earlier version carried $\mu_c = \mu + Sb_c$, with $b_c \sim \mathcal{N}(0, \psi^2 I_q)$ and S a fixed $P \times q$ loading. It was removed.

The construct itself is ordinary: a cohort random effect is the second level of any NLME model. The failure was S . Its columns must say which directions in parameter space a cohort may move along, and no evidence specifies them. Choosing them by hand encodes a guess as a structural constraint. Choosing $S = I_P$ adds a second variance component that competes with ω and that these data cannot resolve.

The decisive objection is one of consistency. Section 11 deletes the basis matrix W , and the reason given is that a low-rank loading in parameter space, whose columns we chose rather than measured, is not a biological statement. S is the same kind of object and it fails the same test. It was introduced three sections after its twin was rejected.

The identification was thin in any case. A cohort holds about two observables, so two rows had to separate a q -dimensional mechanistic shift from a free δ_m per observable plus a shared η_c . Removing b_c , S , ψ and q costs nothing that was ever identified. The covariate model above recovers the same intent with coefficients that pool, and the eligibility rule recovers part of it for free.

8.5 The repeated endpoint

One endpoint does appear more than once. The pooled CD8 target combines three cohorts into one row. Split it back out. This is the single most valuable change in this document, and it is extraction work rather than modelling work.

Write m^* for that observable and c_1, c_2, c_3 for its cohorts. The three centre rows are

$$\tilde{Q}_{m^*, c_k}^{\text{ctr}} = Q_{m^*}^{\text{ctr}}(\phi) + \eta_{c_k} + \delta_{m^*}, \quad k = 1, 2, 3. \quad (18)$$

The three rows share one δ_{m^*} and differ only in η . Four things follow.

Two contrasts of η become identified. The differences $\eta_{c_1} - \eta_{c_2}$ and $\eta_{c_1} - \eta_{c_3}$ are read directly from the data. Only the mean of the three stays aliased with δ_{m^*} . This is the first identified study effect in the model.

τ_η gets one direct measurement. Two contrasts are a weak estimate of a scale, and they are the only estimate available. τ_η then sets how much of every *other* residual in the model goes to a study effect rather than to a model error. The value of these two contrasts therefore reaches far past their own three rows.

The pooled width was too large. A pool of three cohorts with different centres is a mixture. Its interquartile range holds the between-patient spread *and* the spread of the three centres. The pooled row therefore fed ω with a width that no single cohort has. The shortfall figure of section 8.6, 0.62 of the observed interquartile range, is biased by this in the direction that makes the simulator look narrower than it is. Recompute it after the split.

The split is a variance-components decomposition. Within a sub-cohort the width is between-patient variability. Across the three sub-cohorts the medians differ by a study effect plus a sampling error. This is the one place in the whole dataset where both quantities are visible for the same endpoint. It is also a direct test of Σ_{samp} : the disagreement of the three medians must match what the bootstrap of section 6.1 predicts.

Two conditions apply. The sources must report a per-cohort n , median and interquartile range, not only the pool. And the three sub-cohorts are small, so the asymptotic quantile covariance would have been a poor description of them. The bootstrap form of Σ_{samp} is what makes the split affordable.

If the three sub-cohorts report a centre only, then $A = 116$ rather than 118, and the width argument above still holds for the pooled row that they replace.

Search the remaining sources for the same pattern. Every further repeated endpoint moves section 12 further from a prior structure and closer to a measurement.

One further question is worth asking of these three cohorts alone. Do any of them report a second observable? If they do, then the three cohorts hold a repeated endpoint *and* more than one readout, which is the only configuration in the whole set that could separate an assay effect from a recruitment effect. That is the condition in assumption (vi), and this is the one place where it might already be satisfied. Check it before accepting that assumption as unavoidable.

8.6 What separates a narrow model from a narrow population

Both κ and ω can widen a predicted interquartile range. The rows in (7) make the competition exact and one dimensional, so we can state it rather than argue about it.

To first order the variance of observable m across patients is

$$\text{Var}[x_m] \approx \sum_{j,k} J_{mj} J_{mk} R_{jk} \omega_j \omega_k, \quad J_m = \partial g_m / \partial \vartheta. \quad (19)$$

Substitute (14). Every assumed width carries the same factor e^s . Write λ_m for the share of that variance that comes from the measured set $\mathcal{M}$. Then

$$Q_m^{\text{wid}} \approx \text{const}_m + (1 - \lambda_m) s, \quad \text{so} \quad \tilde{Q}_m^{\text{wid}} \approx \text{const}_m + (1 - \lambda_m) s + \log \kappa_0 + \tau_\kappa \kappa_{\text{raw},m}. \quad (20)$$

Three statements follow, and each one is checkable.

If $\lambda_m = 0$ for every observable, then s and $\log \kappa_0$ are aliased exactly, to first order. Only their sum appears. The prior then decides the split, and nothing else does. This is the honest reading of the current model.

The break is λ_m , and it must vary across observables. A measured width takes no s , by (13). An observable driven mostly by $\mathcal{M}$ therefore reads $\log \kappa_0$ almost directly, and the difference between a high- λ observable and a low- λ one separates the two. Compute λ_m for all M_w observables and report the spread before you quote the split. With $|\mathcal{M}| = 9$ the values may all be near zero, in which case the split is a prior statement and must be reported as one.

The earlier argument was wrong in a way that mattered. It compared κ_0 against one ω_j and concluded that a uniform shortfall is cheap under κ_0 , because no single parameter feeds every observable. A uniform rescaling of *all* widths does feed every observable, exactly. What separated the two was the prior mass: a common shift of P independent $\log \omega_j$ costs $P(\log c)^2 / 2\tau_\omega^2$, which grows with P . The split therefore depended on how many inert parameters stayed in the vector. Removing 200 unused parameters would have moved it. The form (14) removes that dependence, because the global direction is now one parameter with one prior width τ_s .

The current model produces about 0.62 of the observed interquartile range at the median observable. Under (20) that uniform shortfall lands on the ridge $s + \log \kappa_0$. Report the joint posterior of $(s, \log \kappa_0)$ as a two dimensional figure. A ridge is a result. A marginal that hides a ridge is not.

Recompute the 0.62 after the CD8 split of section 8.5. A pooled cohort reports the width of a mixture, which is wider than the width of any component. Any pooled row therefore pushes this figure down, and it does so for a reason that has nothing to do with the simulator.

A third action attacks the same problem from the data side. With three quantile levels, κ_m copies any width change exactly, because it stretches the prediction about its own median. With five levels, or a reported range, or a reported mean beside a reported median, a change in ω changes the *shape* of the pushforward and κ cannot copy it. Every extra level that a source gives is a direct attack on this aliasing, and it costs nothing but extraction work.

Per observable the split is still not identified. Each observable with a width row has one centre row and one width row. δ_m takes the centre and κ_m takes the width, which leaves nothing. So τ_δ and τ_κ carry the separation, exactly as in section 12.

9 How we build ω_0

The vector ω_0 is the prior centre for the marginal widths in (13) and (14). We build it in four layers. Each layer replaces the layer before it.

L1. Default. $\omega_j = 0.35$ for every parameter. This is a coefficient of variation of about 36%. A parameter gets this value when we know nothing more specific.

- L2. Class.** A geometric packing limit gets 0.15. A material constant gets 0.10. The spread that we observe for these classes is mostly assay noise. To call it ω would overstate the real diversity.
- L3. Explicit.** One value for one parameter, when we hold a specific belief about it.
- L4. Data.** A submodel target can report a spread across biological units. We then combine that spread with the value from L1 to L3. We do not replace it. A parameter that reaches L4 enters the set $\mathcal{M}$.

For L4, the sampling standard deviation of a log spread from n units is about $\{2(n-1)\}^{-1/2}$. The data therefore carry precision $2(n-1)$ against a class-prior precision τ^{-2} . This gives

$$\log \omega_{0j} = \frac{2(n_j - 1) \log \hat{s}_j + \tau^{-2} \log \omega_j^{(0)}}{2(n_j - 1) + \tau^{-2}}, \quad \tau = 0.5. \quad (21)$$

In this equation, $\hat{s}_j$ is the spread that the source reports, n_j is the number of biological units behind it, and $\omega_j^{(0)}$ is the value from L1 to L3. The data become stronger than the prior at about $n = 3$. The precision in the denominator is also the precision that (13) uses as the prior width for $j \in \mathcal{M}$.

A parameter with no value of n_j keeps $\omega_j^{(0)}$. A spread with no sample size carries no information.

What L4 admits. Equation (21) applies only when the source measures variation across biological units, such as patients, donors or animals. It does not apply to technical replicates, to translation between experimental contexts, or to an assumed value. Those are measurement or epistemic quantities and not variability between patients.

This gate matters more than it appears. The submodel population block covers 178 parameters, but for 145 of them the reported spread is numerically equal to the epistemic posterior width. Without the gate, the model would take those 145 epistemic widths as between-patient spread. At present the gate admits 9 of 271 parameters. The other 262 come from L1 to L3.

The gate now does a second job. $\mathcal{M}$ is exactly the set that L4 admits, and section 8.6 needs $\mathcal{M}$ to be non-empty to identify κ_0 at all. Every parameter that the gate admits therefore buys identification as well as a width. Widening the gate to buy identification would defeat it.

10 Where R comes from

R is a separate input to the model. Nothing in the fit changes it, and you can replace it without touching anything else. At present it is the correlation matrix of the stage-1 composite prior,

$$R = \text{Corr}(\text{Cov}_\pi[\vartheta]). \quad (22)$$

The stage-1 prior joins submodel posteriors through a Gaussian copula, so (22) carries whatever structure that fit produced. The copula covers 65 of the 271 parameters. The remaining 206 parameters come from independent marginals, so R states independence for every pair that involves one of them.

What (22) does and does not give. Two measured facts qualify it.

The matrix is close to the identity. Its copula covers 65 of 271 parameters, and only 26 off-diagonal entries exceed 0.5 in absolute value, out of 36,585. For most parameter pairs, (22) states independence.

Where the matrix is not close to the identity, the entry is epistemic. The largest, at -0.994 , joins two rates that one target constrains only through their sum. That number describes what the stage-1 data could not separate. It does not describe how patients differ. A second group of strong entries comes from declared hierarchical parameter groups; those encode a real claim that the members are the same kind of quantity, but their magnitudes are still epistemic.

R is therefore best read as a placeholder with the right interface, not as a measured quantity.

Better candidates for R exist and slot into the same place:

- a block matrix built from the declared parameter groups, with a within-group correlation elicited rather than read from stage 1;
- structural constraints, such as rates that partition a conserved total;
- the identity, if the conclusion is that nothing is known;
- a shrunk form $(1 - \lambda)I + \lambda \text{Corr}(\text{Cov}_\pi[\theta])$.

Because R enters only through (1), a comparison across these choices is a sensitivity analysis and not a redesign. Section 14 lists that comparison, and it has not yet been run.

A correlation is not a constraint. Some parameters partition a conserved total, so their values must satisfy an identity. R cannot state an identity. A correlation of -1 makes two parameters move together on a line, but the line is not the constraint surface, and a lognormal draw does not sit on it. Only a reparameterisation can enforce such a constraint. Section 13 filters the draws that break one, which detects the problem and does not fix it.

11 What the data constrain

The data constrain Ω only through the width of an observable distribution. To first order the spread of observable m across patients is $J_m \Omega J_m^\top$. Each width row therefore gives one linear function of Ω .

Published summaries are marginal, one observable at a time. The cross terms $J_j \Omega J_i^\top$ are never observed. Of the 65 targets, 49 report a usable scale. There are therefore at most 49 constraints on Ω , and κ takes some of them. There are 51 width rows, because the CD8 target of section 8.5 contributes three. Rows and constraints are different counts: those three rows measure the same function of Ω three times, which buys precision and not a new direction.

Two consequences follow.

First, no part of R can be estimated from these data. Its entries are exactly the cross terms that a marginal summary never reports. R must come from outside the fit, which is why section 10 treats it as an input.

Second, most of the 271 widths are not informed either. Along those directions the posterior returns the prior (14). The reported population is then a statement of ω_0 , not a measurement.

Design note: there is no basis matrix. An earlier version replaced (1) by $\vartheta_i = \mu + W(\sigma_u \odot z_i)$. Here $W = L\tilde{V}$ came from an eigendecomposition of the observation information in the metric of the prior covariance, and the fit estimated $K = 15$ direction multipliers instead of 271 widths. The construction is known as the likelihood-informed subspace, or the active subspace.

The rotation and the truncation were removed. The correlations were kept, and they are now R .

Three reasons. First, σ_u was not interpretable: it scaled directions, so no single entry was the width of any parameter. Second, the correlations did not stay fixed. Because $\Omega = \Gamma + \sum_{k \leq K} (\sigma_{u,k}^2 - 1) w_k w_k^\top$, the fit moved $\text{Corr}(\Omega)$ by a rank- K update, and that update lay along the directions that the *measurements* resolve. A correlation that appears because a combination happened to be observable is not a biological statement. Third, the truncation to K directions limited how much model error the fit could report as patient diversity. That is protection by restriction and not by correctness; section 8.6 now names the correct fix.

The present form keeps what the rotation was supposed to deliver, namely the prior correlations, and states them as an explicit and replaceable object.

12 The corrections are aliased

Sixty-four of the 65 observables belong to exactly one cohort. Only the CD8 target of section 8.5 appears in more than one. That fact has a consequence that the notation hides, and it applies to all but one observable.

Take a cohort c that holds observables $m_1, \dots, m_k$. Equation (6) uses only the sum $\eta_c + \delta_{m_i}$. Add a constant to η_c , subtract the same constant from every δ_{m_i} in that cohort, and the likelihood does not change. η and δ are therefore aliased, exactly, and only their priors separate them.

The eligibility rule of section 8.4 is the one cohort device that escapes this. It enters before g and it costs no parameter, so there is nothing for δ to trade against. A cohort effect that carries a free parameter does not escape, whichever level it acts on, which is why section 8.4 removed the parameter version rather than tuning it.

What η is, and is not. In an ordinary meta-analysis, a study effect is identified because the same endpoint appears in several studies. One endpoint does that here, and it gives two contrasts. For the other 26 cohorts η_c is not an identified study effect.

Read those as a correlation structure on the corrections: the model states that the errors within one study share a common part. That claim has content, and it is a reasonable claim, but it is a prior structure and not a measurement.

What changes with the CD8 split is the *scale*. τ_η is now measured rather than asserted, and τ_η decides how much of every other cohort's residual is a study effect. Two contrasts are a thin basis for that, so report the posterior for τ_η beside any statement that depends on it. Section 8.4 names the two routes to more: further repeated endpoints, or a cohort attribute table.

The count is worth stating plainly. The model has 65 values of δ , 49 of κ and 29 of η , against $A = 118$ anchor rows. Sixteen observables have a single anchor row each, because their source

reports a centre only; for those, δ_m absorbs the residual exactly.

Therefore μ is not identified by any single observable. It is identified by the *agreement* of residuals across observables, through the shrinkage of δ toward zero. The same holds for ω through κ . The scales τ_δ and τ_κ do the work, and they are estimated from how much the corrections disagree with each other.

This is a partial-pooling argument and it is sound. But the model does not have 118 independent measurements of the population, and no report should suggest that it does.

13 How to emit the population

The fit gives a posterior over ϕ . A virtual population is a set of patients. The rule that turns one into the other is part of the model, and getting it wrong throws away the distinction that section 8.1 was careful to make.

The object that carries both quantities is the posterior predictive over patients:

$$p(\theta_{\text{new}} \mid \text{data}) = \int F(\theta \mid \phi) p(\phi \mid \text{data}) d\phi. \quad (23)$$

Its spread holds the variability between patients *and* our uncertainty about the population. Do not collapse the two by accident. We emit three products, and each one carries its own label.

The reference population. Draw at one $\hat{\phi}$: the posterior mean of μ and the posterior median of ω . Its spread is between-patient variability only. Use it when one population must be shipped, and say on the label that it carries no uncertainty about ϕ .

The ensemble. Keep S populations, one per posterior draw of ϕ . Run the downstream study once per population. Report the spread of the *results* across populations. This keeps the two variances separate in the answer, and it is the default deliverable.

The pooled predictive. Equation (23) itself: draw $\phi^{(s)}$, then draw patients from $F(\cdot \mid \phi^{(s)})$, then pool. It is wider than the reference population. Use it only when the decision needs one population that is right on average.

Four rules apply to all three.

1. **Draw fresh z .** The N frozen patients are a device that makes the gradient work. They are not a population. Draw new z for the deliverable, at whatever size the deliverable needs.
2. **Carry the corrections forward.** For an observable that the fit used, apply δ_m and κ_m to the prediction. For a *new* observable, draw $\delta_{\text{new}} \sim \mathcal{N}(0, \tau_\delta^2)$ and $\log \kappa_{\text{new}} \sim \mathcal{N}(\log \kappa_0, \tau_\kappa^2)$ from the fitted hyperparameters. This is what the hierarchical form is for. A prediction that drops the corrections claims that the model is correct, after a fit that measured how wrong it is.
3. **Filter for viability, and report the rate.** Score each drawn θ with the restriction classifier (`inference/restriction.py`), which predicts whether the simulator returns a usable output. Run the survivors through the full simulator, not the emulator. Report the rejection rate. A high rate means that F puts mass outside the feasible set of the model. That is a fault in the fit and not a step in post-processing.
4. **Keep prevalence weighting for one job only.** If a downstream user needs the population to match a reported marginal exactly, reweight the emitted patients (`vpop/weighting.py`, Allen et al. 2016). Report that as an adjustment on top of the fit. It is never the fit.

14 Checks

No number in this model is reportable until the checks below have run. They run in this order. Each one gates the next, because a later verdict means nothing while an earlier one fails.

1. **Simulation-based calibration on ϕ .** Draw ϕ^* from the prior, simulate anchor rows through (8), refit, and check that ϕ^* lands uniformly inside its own posterior. This tests the sampler and the likelihood code. It never sees the real data. If it fails, nothing below is readable.
2. **The width split.** Generate synthetic data with a known ω and a known uniform narrowing of the simulator. Fit. Confirm that s takes the narrowing and u takes the pattern. Report the recovered $(s, \log \kappa_0)$ pair against the truth. Report λ_m for all M_w observables beside it.
3. **The frozen cloud.** Run the whole fit three times with three different frozen clouds $z_{1:N}$. Compare the posteriors. The difference is the bias that Σ_{MC} hedges and does not remove.
4. **The plug-in for Σ .** Change ϕ_0 by a factor of two and refit. Section 6.1 records the result of this test on the earlier form of Σ . Repeat it on the bootstrap form.
5. **The three CD8 cohorts.** Their three medians differ by a study effect plus a sampling error. Check that the disagreement matches what the bootstrap Σ_{samp} predicts. This is a direct test of section 6.1, and it is the only one available. Report the posterior for τ_η that these two contrasts produce, and compare it against its prior.
6. **Standardised residuals.** Compute $(y - \tilde{Q})_r / \sqrt{\Sigma_{rr}}$ for all A rows. Look for structure by cohort, by observable class and by n_c . Structure by cohort means that the cohort terms of section 8.4 are doing the wrong job.
7. **Leave one cohort out.** Refit without cohort c , then predict its rows and check coverage. Repeat over cohorts. This is the only external check in the list, and it is the one that tests whether τ_δ and τ_κ are honest.
8. **The emulator, where the fit reads it.** Section 5. Draw from $F(\theta \mid \hat{\phi})$, run the full simulator, and compare against the emulator. Report the result beside the design residual.
9. **The variance budget.** For every parameter, report which of three things produced its width: the data moved it, the prior returned it, or the pool could not represent it. Quote no per-parameter width without this table.
10. **Sensitivity to R .** Refit under the four candidates of section 10, including the identity. Report how far $\hat{\mu}$ and $\hat{\omega}$ move. R is assumed, so this number is part of the result.
11. **The viability rate.** Section 13, rule 3. Report the fraction of emitted patients that the simulator rejects.

15 What the fit estimates

Quantity	Status	Source
μ	estimated, P	Prior (12) from the stage-1 mean and covariance.
s	estimated, 1	The global width multiplier. Aliased with κ_0 ; see section 8.6.
u	estimated, $P - \mathcal{M} - 1$	The pattern of the widths. Sums to zero.
$\log \omega_{\mathcal{M}}$	estimated, 9	Measured widths. Prior width from the source sample size.
η, δ	estimated	Centre corrections. Aliased; see section 12.
κ, κ_0	estimated	Width corrections. See section 8.6.
$\tau_{\eta}, \tau_{\delta}, \tau_{\kappa}$	estimated	These carry the separation of the corrections.
ε	fixed	Emulator residual, on the diagonal of Σ . Section 5.
$R = \text{Corr}(\Omega)$	fixed	Section 10. An input, not a result.
ω_0	fixed	Section 9.
$\mathcal{E}_c$	fixed	Eligibility rules. Section 8.4.
Σ	fixed	Section 6.1, at a plug-in ϕ_0 .
$z_{1:N}$	fixed	Drawn one time.

The fit estimates about 690 quantities against $A = 118$ anchor rows. At most 49 functions of Ω are identified by the data, so the prior carries most of ω , and the shrinkage of the corrections carries the rest. A count of free parameters is not a warning by itself in a hierarchical model. It is a warning here, because section 12 shows that the 118 rows are not 118 independent measurements.

16 Assumptions

- (i) **We assume R rather than estimating it.** The correlations of the population are an input (22), and no part of them is identified by these data. The current choice is the stage-1 correlation matrix, which is epistemic: it shows how our uncertainties move together, not how patients differ. It is also close to the identity. Section 10 lists better candidates, and check 9 measures the sensitivity. That check has not yet been run.
- (ii) **No data gives the correlations between patients.** Papers report each observable on its own, so the cross terms $J_j \Omega J_l^T$ are never observed. To measure them, we need two or more observables in the same individuals. Digitised per-patient figures could supply this, but only if the extraction keeps the identity of each patient across panels.
- (iii) **The spread is mostly prior, not data.** At most 49 functions of Ω are identified against 271 free widths. Also, 262 of the 271 values of ω_0 come from a class default rather than a measurement. The width of the reported population is therefore in large part a statement of assumption. This is why ω_0 must come from external evidence, and why τ_{ω} should not be widened for convenience.
- (iv) **The global width and the global model error share one direction.** Equation (20) shows that s and $\log \kappa_0$ enter every width row through the sum $(1 - \lambda_m)s + \log \kappa_0$. If λ_m is near zero for every observable, the prior width τ_s decides the split and the data do not, because s and $\log \kappa_0$ share it. Report λ_m and the joint posterior of the pair. Check 2 tests the split on synthetic data, and it has not yet been run.
- (v) **The centre corrections are aliased and the priors separate them.** Section 12 shows that η_c and δ_m are exactly aliased for 64 of the 65 observables, because those observables appear in one cohort each. The count of anchor rows overstates the information by a wide margin. Any statement about μ or ω depends on τ_{δ} and τ_{κ} remaining small, which is an assumption about how consistent the sources are.

The CD8 split of section 8.5 is the one exception, and it changes the status of τ_η from asserted to measured. It rests on two contrasts. Treat it as one weak measurement and not as a solution.

- (vi) **Cohort differences are assumed to be assay effects.** η_c shifts every readout of a study by a common amount, which is a statement about calibration. A cohort that recruited different patients would instead shift the parameters. For a cohort with one observable the two predict the same number, and most cohorts have about two, so these data cannot separate them. The model takes the assay reading by construction, and every statement about μ inherits that choice. Section 8.4 explains why we removed the parameter-level term rather than carrying both.
- (vii) **The emulator error term is specified and not yet populated.** The slot ε_m enters Σ through (9). In the current pipeline nothing writes to it, so it is zero. The emulator's held-out residual is not negligible: relative to the within-cohort spread it lies between 0.095 and 0.254. Until ε is wired, every target is trusted more than it should be, and the highest- n targets most of all. Section 5 also requires that this residual be re-measured on draws from the fitted population, and that has not been done.
- (viii) **The measured spreads are biased in both directions.** A reported across-patient spread also contains assay noise and biopsy-site variation, which makes it too large. A spread measured across animals of one genetic background is smaller than a spread across human patients, which makes it too small. We correct neither effect in (21).
- (ix) **Parts of this note describe the model and not the code.** The document is the specification. At the time of writing, `vpop/population_fit.py` still uses the basis form $\vartheta = \mu + W(\sigma_u \odot z)$ and has no κ , no s and no u , no eligibility rule, and an asymptotic Σ . The items to build are: the $(s, u, \mathcal{M})$ split of section 8.2; the two-row anchors of (5); the additive $\log \kappa$ of (7); the bootstrap Σ_{samp} and the measured Σ_{MC} of section 6.1; the cohort terms of section 8.4; the stage-1 covariance in (12); and the emission rules of section 13. The CD8 split of section 8.5 is also outstanding, and every count in this note that involves C or A assumes it. Do that one first. It is extraction work, it needs no new code path beyond a cohort index, and it is the only change here that adds information rather than structure.

17 What data to gather next

Section 16 lists nine assumptions. Most of them are not defects of the method. They are statements that the available data cannot answer a question. This section says which data would answer which one, and in what order to go and get it.

The rule that sets the order. A new anchor row is worth what it shares with a row that you already hold.

A row that shares an *endpoint* with another study identifies η . A row that shares a *cohort* with another observable separates an assay effect from a recruitment effect, and helps μ . A row that shares a *patient* with another observable identifies R . A row that shares nothing arrives with its own δ_m and its own κ_m , so it contributes only through the shrinkage of the corrections.

This rule reverses the usual instinct. One more observable, in one more cohort, at the same three quantile levels, is the least valuable thing that you can add. It is also the easiest thing to add.

17.1 Price the options before you buy them

Three checks in section 14 cost almost nothing, and they set the ranking below. Run them first.

Check 9, the sensitivity to R . Refit under the identity and under the stage-1 matrix. If $\hat{\mu}$ and $\hat{\omega}$ barely move, then item 1 below is worth less than it claims. If they move, item 1 is the whole question. You cannot know which without the run.

The λ_m report of section 8.6. It says whether measured parameter spreads can break the ridge between s and $\log \kappa_0$, and which observables would do it.

Check 8 and the emulator resolution of section 5. A parameter that the emulator cannot resolve gains nothing from a better prior. Do not buy data for those parameters.

17.2 The ranking

1. **Two or more observables in the same patients.** This is the only source of R , and it is the only item that attacks an assumption that section 11 calls unreachable from summary data. It does four jobs at once. It gives the cross terms $J_j \Omega J_l^T$. It gives a full marginal instead of three quantiles. It puts several observables in one cohort. It removes the plug-in from Σ .

The cheap form of this item exists today. Papers in this area print per-patient dot plots. Digitise them, and keep the identity of each patient across panels. Twenty patients with five observables each would change this model more than twenty new targets. The extraction discipline is the whole requirement. A panel digitised without patient identity is worth a fraction of the same panel digitised with it. Removes (i) and (ii).

2. **Repeated endpoints across studies.** The cheapest item here, and the CD8 target of section 8.5 proves that the stock exists. For each of the 65 observables, search for a second publication that reports the same quantity in a comparable cohort. Five repeated endpoints move section 12 from a prior structure to a measurement, and τ_η then governs how the model splits every other residual between a study effect and a model error. Removes (v).
3. **Across-patient spreads for parameters, ordered by sensitivity.** Nine of 271 parameters have a measured spread. Each one that you add does two jobs: it improves ω_0 , and it enters $\mathcal{M}$, which is what makes λ_m vary and breaks the width ridge. The second job is why the early returns are large.

Do not search at random. Rank the parameters by their contribution to the predicted variance of the observables that have a width row. That ranking is computable today from J_m , R and ω_0 . Then find donor-to-donor or patient-to-patient spreads for the top of the list, with sample sizes. The gate of section 9 still applies: biological units only, and never replicates. Removes part of (iii) and part of (iv).

4. **More quantile levels from the papers that you already cite.** With three levels, κ_m copies any width change exactly. With a reported range, a tenth and a ninetieth centile, or a mean printed beside a median, it cannot, because ω changes the shape of the pushforward and κ only stretches it. This is a re-read of sources that are already in the set. It is the cheapest attack on the largest remaining ambiguity. Removes part of (iv).
5. **Cohort attributes and eligibility criteria.** For all 29 cohorts: stage, line of therapy, treatment status, sample type, assay platform, region. This turns 29 free values of η_c into $Z_c \beta$, whose coefficients pool across cohorts, and it is also what a covariate model on the parameters would need. A quantitative eligibility rule is better still, because it removes

a parameter instead of constraining one. This is data entry against sources that we hold. Removes part of (v) and part of (vi).

The attribute table also does one thing that no other item does. An attribute that is plainly instrumental, such as the assay platform, and an attribute that is plainly clinical, such as the stage, load on the same η_c today. Split them and the assay reading of assumption (vi) becomes testable rather than asserted.

6. **Repeat measurements on the same patient.** Assumption (viii) says that every reported across-patient spread also holds assay noise, and that we correct none of it. So every ω_0 that L4 built is too large by an unknown amount. A technical replicate, or a paired sample from one patient, measures that noise directly and lets us subtract it. This is the only item in the list that removes a known bias rather than a known ambiguity. Removes (viii).

17.3 Summary

Data	Removes	Cost
Per-patient values, several observables, identity kept	(i), (ii)	digitisation
The same endpoint in a second study	(v)	literature search
Parameter spreads with n , ranked by sensitivity	(iii), (iv)	literature search
Extra quantile levels from current sources	(iv)	re-read
Cohort attributes and eligibility rules	(v), (vi)	data entry
Replicate or paired samples	(viii)	new measurement

17.4 If we can commission one measurement

One cohort. One platform. Five to ten of the model's observables, measured in the same patients, with the per-patient values published. That is item 1, item 5 and part of item 2 in a single study. Nothing else on the list gives three at once.

17.5 What not to buy

A new observable, in a new cohort, at three quantile levels. It arrives with a new η_c , a new δ_m and a new κ_m . The model gains one row and three parameters, and the information reaches μ and ω only through τ_δ and τ_κ .

The same observable, placed in a cohort that is already in the set, is worth several times more. It adds no η , and it gives that cohort a second observable, which is what assumption (vi) needs before an assay effect and a recruitment effect can be told apart. When there is a choice of source, that is the choice.

17.6 Make this list computable

`inference/obed.py` ranks candidate measurements by expected information gain. Point it at $(\mu, s, \log \kappa_0, \log \omega_{\mathcal{M}})$ rather than at the full ϕ , and this section becomes a number instead of a judgement. The ranking above is what we expect it to return. The run is how we find out where the expectation is wrong.